Exercise Sheet 2

[DATA23UF] Probability and Statistics for Data Science

October 29, 2026

1. 🌶🌶🌶 Let (X, Y) be a random vector with values in $\{0,1\}^2$ and joint probability mass function

$$p_{X,Y}(x,y) = c(1+x+y), \qquad (x,y) \in \{0,1\}^2,$$

where $c > 0$ is unknown.

   a) Determine c.

   b) Determine the marginal probability mass functions p_X and p_Y. Are X and Y independent? Justify your answer.

   c) Define $Z := X + Y$. Determine the probability mass function of Z.

   d) Compute $\mathbb{E}[X]$, $\mathbb{E}[Y]$ and $\mathbb{C}\mathrm{ov}(X,Y)$.

2. 🌶🌶🌶 Let $X_1, X_2, \ldots \overset{\text{iid}}{\sim} \mathrm{Ber}(p)$, $p \in (0,1)$, and let $r \in \mathbb{N}$. Define

$$T_r := \min\Big\{n \in \mathbb{N} : \sum_{i=1}^{n} X_i = r\Big\}.$$

Thus T_r is the number of experiments required to observe the r-th success.

   a) Let $n \geq r$. Show that

$$\mathbb{P}[T_r = n] = \binom{n-1}{r-1} p^r (1-p)^{n-r}, \qquad n = r, r+1, \ldots.$$

   b) Define $N_r := T_r - r$. Interpret N_r and determine its probability mass function.

3. 🌶🌶 Let $X, Y \overset{\text{iid}}{\sim} U[0,1]$ and define $S := X + Y$.

   a) Use the convolution formula to show that

$$f_S(s) = \begin{cases} s, & 0 \le s \le 1, \\ 2 - s, & 1 < s \le 2, \\ 0, & \text{otherwise}, \end{cases}$$

   and determine the cumulative distribution function F_S.

   b) Compute $\mathbb{E}[X]$ and $\mathbb{V}\text{ar}(X)$.

   c) Determine $\mathbb{E}[S]$ and $\mathbb{V}\text{ar}(S)$ in two different ways, once using the representation $S = X + Y$ and once directly from the density f_S.

4. 🌶🌶🌶 Let $\mathbf{X} \sim \mathcal{N}_d(\boldsymbol{\mu}, \Sigma)$, where $\Sigma \in \mathbb{R}^{d \times d}$ is symmetric and positive definite. Let $A \in \mathbb{R}^{d \times d}$ be invertible and $\mathbf{b} \in \mathbb{R}^d$, and define

$$\mathbf{Y} := A\mathbf{X} + \mathbf{b}.$$

   a) For $g(\mathbf{x}) := A\mathbf{x} + \mathbf{b}$, determine the inverse transformation g^{-1} and the Jacobian determinant

$$\left| \det Dg^{-1}(\mathbf{y}) \right|.$$

   b) Show that

$$\mathbf{Y} \sim \mathcal{N}_d\Big(A\boldsymbol{\mu} + \mathbf{b}, A\Sigma A^T\Big).$$